

Design and Evaluation of Machine Learning Workflow for Breast Cancer Diagnosis Classification

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Abstract

The accurate diagnosis of breast mass as malignant or benign is a critical step that enables the timely treatment. It can also reduce the need of unnecessary biopsies. For this purpose, this reproducible machine learning project is performed by using the Winconsin Diagnostic Breast Cancer dataset (WDBC) from UCI Machine Learning Repository. Data analysis, correlation and redundancy analysis, feature selection and comparison of seven supervised classification algorithms are covered. GridSearchCV is used to tune the algorithms. 20 percent of the dataset is used to test the metrics accuracy, F1 score, recall and precision, MCC, ROC and AUC. SelectKBest and Recursive Feature Elimination is used to identify the subset of parameters that should be used to have a solid model. In the end, common 16 features out of 30 is selected. Logistic regression had the best test-set performance, having 98.2 percent accuracy, 0.96 MCC, and 0.998 AUC. PCA projection also showed that the separation can be done linearly; thus, aligning with the result of non-linear models performed worse than Logistic Regression.

Introduction

Breast cancer is one of the most diagnosed cancers. Early treatment and accurate diagnosis have a direct effect on the outcomes of the treatment and survival. Visual and manual interpretation of fine needle aspirate biopsies are used traditionally to have information about the mass. However, this procedure can be time consuming and depending on the observed, the diagnosis can differ (Adedayo-Ajayi et al., 2023). With the availability of digitalized cytological measurements, statistical classification methods are starting to become more common to support traditional methods (Casaubon et al., 2026). Using machine learning for this type

of diagnostics is particularly suited since the decision boundary between malignant and benign cells is described by several geometric measurements rather than a single threshold (Sharma et al. 2018). The dataset is Wisconsin Diagnostic Breast Cancer dataset, suitable for binary classification used by machine learning workflow (Wolberg et al., 1993).

Examining how and why certain algorithms perform better than others that are listed is one of the aims of this project in addition to the identifying the best performing classifier for this dataset given the structure of data. Also, smaller subset of features will be tested to match the performance of 30 feature set.

Materials and Methods

Dataset

The dataset taken from UCI Machine Learning Repository is Wisconsin Diagnostic Breast Cancer (WDBC) dataset. It has 569 digitized images of fine needle aspirate of breast masses. All samples have 30 numeric features describing the images.

There are 10 cell-nucleus properties: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension; and each of them are summarized as its mean value, standart error, and worst value, making the features 30 in total.

Diagnosis is labelled as 1 and 0, meaning malignant for 1 and benign for 0. There is no missing values in the dataset, indicated by the dataset owners as well as checked in the workflow I performed. Thus, no imputation was required.

Exploratory Data Analysis

Before performing any modeling, data analysis is performed. The analysis includes descriptive statistics, class distribution plots, histograms, density plots, box-and whisker plots separated by diagnosis, Pearson correlation heatmap, and scatter matrix. Throughout the analysis, shape, scale and redundancy of the feature set is examined.

The dataset's class distribution is shown in Figure 1. It is not balanced, with 357 benign and 212 malignant labels. Inclusion of metrics addition to the raw accuracy is motivated such

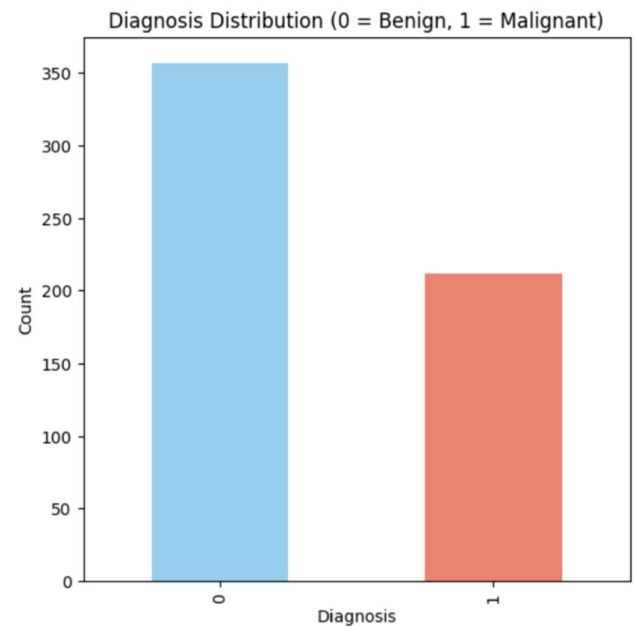


Figure 1. Class distribution of the dataset

as F1 and MCC to have clear results regardless of the imbalance in the dataset.

Descriptive statistics also show that feature set includes very different numeric scales for different features. As an example, area_mean ranges into hundreds and thousands whereas smoothness_mean is on the order of 0.1. This difference in the scale indicates the necessity of standardization of the feature scales before applying any algorithm.

Correlation Analysis

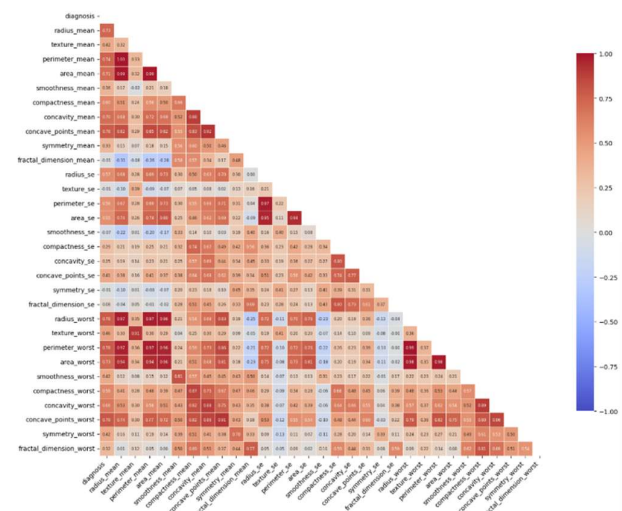


Figure 2. Correlation Matrix for mean-value features

Since the 10 underlying properties measured in three ways and became 30 features, it was expected to see a strong correlation. In Figure 2, it is clearly confirmed that radius, area and perimeter are strongly correlated with each other with large than 0.95 correlation coefficient. This was expected since they are different mathematical descriptions of the same quantity in geometry. Also, concavity and concave points showed a close relationship. This redundancy in the dataset can inflate the variance in the coefficient for linear models and make individual coefficients harder to interpret; thus, this figure is an important consideration for the downstream processing, most importantly, for feature selection.

Train/Test Split and Feature Scaling

The data was split as 80 percent training set and 20 percent test set before feature selection and model fitting steps. I used stratified sampling with `random_state=42` to preserve class ratio in the sets. A `StandardScaler` is fit on the training data only, after that applied to both training and test sets to not have any data leakage. This standardization step is important since the algorithms would react to features being different scales.

Feature Selection

Two feature selection techniques were evaluated by using cross-validated accuracy of a Logistic Regression model to choose the subset of features that are needed to be kept

rather than fixing an arbitrary number without any logic behind.

Univariate Selection (`SelectKBest` with ANOVA F-statistic) selected 19 features and had a cross validated accuracy of 97.58 percent. Recursive Feature Elimination selected 17 features with a cross validated accuracy of 97.58 percent.

16 features (the ones that are intersecting in both selection methods) are considered the most informative: `area_mean`, `area_se`, `area_worst`, `concave_points_mean`, `concave_points_worst`, `concavity_mean`, `concavity_worst`, `perimeter_se`, `perimeter_worst`, `radius_mean`, `radius_se`, `radius_worst`, `smoothness_worst`, `symmetry_worst`, `texture_mean`, `texture_worst`.

PCA

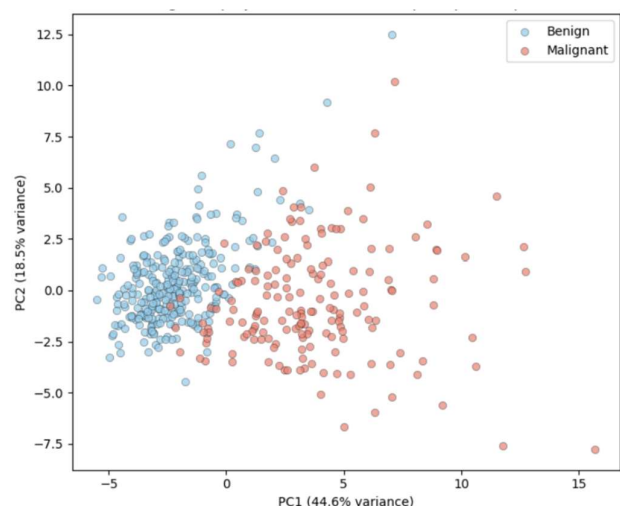


Figure 3. Training data projected onto the first two principal components

PCA is applied to the training data which is scaled, clearly showing the linear separability of the benign and malignant classes in Figure 3. Even though PCA is not directly used for modelling, it was a visual check showing that

the classification problem is close to linear separation.

Model Training and Evaluation

Representing linear, probabilistic, distance-based and tree based machine learning strategies, seven supervised algorithms were compared in this section: Logistic Regression, Linear Discriminant Analysis, k-nearest neighbors, Support Vector Machine, Gaussian Naïve Bayes, Decision Tree, and Random Forest. Six of these algorithms were tuned by using GridSearchCV with 5 cross validation folds on the training set. Gaussian Naïve Bayes is discarded in this step since it does not have relevant hyperparameters. Also, since support vector machine has the largest hyperparameter grid, GridSearchCV was additionally compared against RandomizedSearchCV. Each model was tested on the 20 percent test set. Accuracy, F1 score, precision, recall, MCC, log loss, and AUC is obtained for each of them. Additionally, confusion matrices and classification reports were obtained to evaluate class specific performance of all.

Results

Feature Selection

When the full 30 features set is compared against 16 features subset, the full set obtained 97.58 accuracy, while the reduced set obtained 97.14 accuracy. Considering the subset is almost half size of the full set and is more interpretable, using smaller set is applicable. The subset also requires much

less computational cost without changing how the model works much.

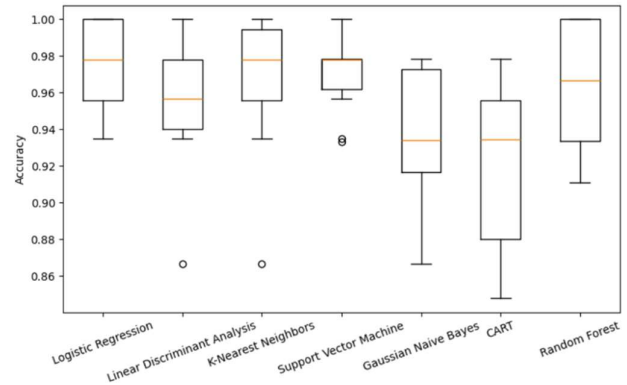


Figure 4. Comparison of 10-fold cross validation accuracy scores with default hyperparameters across seven algorithms

Comparison of Machine Learning Algorithms

As seen in Figure 4, the highest accuracy among the models is obtained by Logistic Regression. Support vector machine is the second, and k-nearest neighbors is the following. The fact that Logistic regression, LDA and SVM (linear models) performed at least as well as non-linear models that are more flexible such as Random forest and CART shows that observation obtained by PCA supports that this dataset is close to linear separation.

Final Tuned Model Performance

After hyperparameter tuning by using GridSearchCV and evaluation on the test set, the results are as follows:

	Accuracy	F1 Score	Precision	Recall	MCC	AUC
Logistic Regression	0.982456	0.975610	1.000000	0.952381	0.962622	0.997685
Support Vector Machine	0.973684	0.962963	1.000000	0.928571	0.944155	0.992725
Linear Discriminant Analysis	0.964912	0.950000	1.000000	0.904762	0.925820	0.997024
Random Forest	0.964912	0.950000	1.000000	0.904762	0.925820	0.996362
CART	0.956140	0.936709	1.000000	0.880952	0.907605	0.939649
K-Nearest Neighbors	0.938596	0.911392	0.972973	0.857143	0.868766	0.982474
Gaussian Naive Bayes	0.921053	0.888889	0.923077	0.857143	0.829162	0.989087

Logistic regression, tuned to $C=0.1$ with an L2 penalty and liblinear solver, produced the best overall test-set result. The comparison between GridSearchCV and RandomizedSearchCV on the SVM's hyperparameter grid showed that both approaches converged to the same configuration: $C=10$, $\gamma=\text{auto}$, $\text{kernel}=\text{rbf}$.

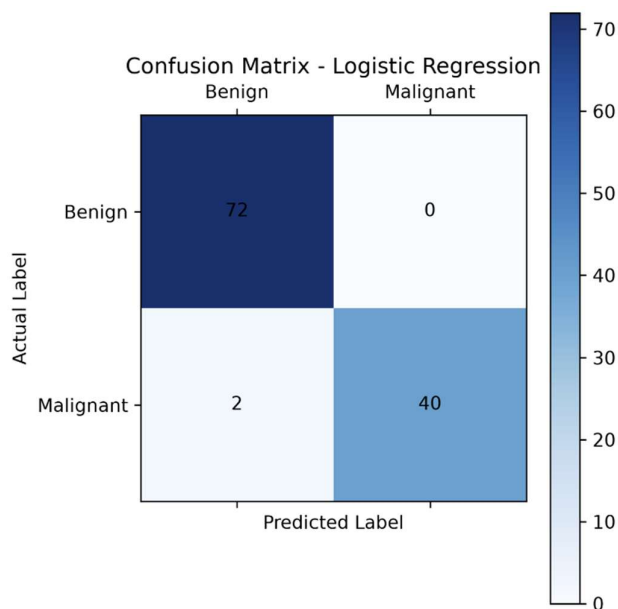


Figure 5. Confusion matrix of the final Logistic Regression model on the held-out

Figure 5 shows that 2 cases out of 114 are misclassified (malignant cases are predicted as benign). This is worth noting since in a clinical case this would affect the diagnosis in a negative way and mislead the patient to not get any treatment. Thus; despite very strong performance of the model, in real life scenario, small number of missed malignant cases carry a higher clinical cost.

Conclusion

This study is performed to reproduce a machine learning pipeline to classify breast cancer masses into two: benign and

malignant. WDBC dataset was used. Data analysis showed that there is a mild imbalance in two classes and a redundancy among 30 features. Feature selection by using two tools agreed on a common 16 features subset, and cross validated accuracy on that reduced set closely matches the full set. This showed that smaller and more interpretable subset is a reasonable choice for further processing. PCA showed that two diagnosis classes are close to be separated linearly. After that, evaluation of the algorithms also showed that Logistic regression performs the best among seven algorithms with different approaches.

The dataset is relatively small and coming from a single source, may be resulting the model to be too fixed on the specific data. An independent benchmark set would be useful to generalize the model further.

Data Availability

All figures, graphs, models and findings as well as the pipeline ipynb file is available on Github: <https://github.com/elif-guler/breast-cancer-ml>

Please check the README page and follow the usage guide to replicate the results.

References

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